Name _____ Date _____ Period _____

Drawing With and Without Replacement

Your bag has **pink**, **orange**, and **red** blocks. Every bag is different, so count yours first.

Pink: _____ Orange: _____ Red: _____ Total: _____

Each trial, draw **2 blocks, one at a time**, without looking.
With replacement: draw one, record its color, **put it back**, shake, draw again.
Without replacement: draw one, record its color, **keep it out**, draw again.
After every trial, put all blocks back in the bag.

Part 1: Predict

1. On the first draw, find: P(pink) = _____ P(orange) = _____ P(red) = _____
What do these add up to? _____

2. Suppose the first block is pink. What is the probability the second block is also pink. . .
if you put the first block back? _____ if you keep the first block out? _____

3. Will getting **two blocks of the same color** be more likely **with** or **without** replacement?
Circle one: With Without Same Why?

_____

_____

Part 2: Experiment

With a partner, do **10 trials** of each method. Tally each outcome (first color, second color).

	With replacement		**Without replacement**	
Outcome	Tally	Count	Tally	Count
P, P				
P, O				
P, R				
O, P				
O, O				
O, R				
R, P				
R, O				
R, R				
Total		10		10

Part 3: Tree Diagrams

Use P, O, and R for the colors.

4. Draw a tree diagram for two draws **with replacement** from your bag. Find the probability of each outcome.

5. Draw a tree diagram for two draws **without replacement** from your bag. Find the probability of each outcome.

6. Use your tallies from Part 2 to complete the table. Give each result as a count out of 10.

Event	**With replacement**	**Without replacement**
Both pink		
Both the same color		
Two different colors		
At least one red		

7. Which event changed the most between the two methods? Explain why, using what happens inside the bag.

8. Find a group whose bag has a different mix of blocks. Did replacing versus not replacing make a bigger or smaller difference for their bag than for yours? Why?

9. A giant bag has 300 pink, 200 orange, and 100 red blocks. Find P(both pink) with and without replacement, to 4 decimal places. Why are these so close when your bag's answers were not?

Part 4: Practice

10. A basketball player shoots two free throws. The probability she **makes the first is 0.7**. If she **makes the first**, the probability she **makes the second is 0.8**. If she **misses the first**, the probability she **makes the second is 0.55**.

(a) Draw a tree diagram to represent this scenario.

(b) Find the probability that she **misses the first** and **makes the second**.

(c) Find the probability that she **makes exactly one**.

11. A bag contains **12** marbles. **Five** are **purple**, **4** are **pink**, and **3** are **green**. Two marbles are drawn at random **without replacement**. Find the probability that:

(a) **both** are **purple**.

(b) one is **purple** and the other is **green**.

(c) both are the **same color**.

(d) **both** are **purple**, if the first marble is **put back** before the second draw.